

BIOGLOW Team Packets

FIRST LEGO League Challenge · BIOGLOW (2026–27) · Grades 3–6 · 30 full packets (forest/urban/rural) · 30 quick starters · 4 cross-tier flagships

FIRST® CANOPY

FLL **BIOGLOW**

FTC **BIOBUZZ**

FRC **BIOCORE**

The spine that connects everything. Identify a problem that puts biodiversity at risk and design a solution. In the rainforest, “countless species depend on one another to survive.” Every idea here is one broken **vital connection** and how to restore it — the CANOPY idea that runs through FLL **BIOGLOW**, FTC **BIOBUZZ** (pollinators) and FRC **BIOCORE** (the core web). This booklet has three parts: **Part 1** ten forest full packets; **Part 2** the same ten jobs as quick starters in each of three settings — **forest, urban, rural** — with the correlation map; **Part 3** the cross-tier connectors that link all three.

How to run any packet (read this first)

These are **starters** — the Innovation Project must be the team’s own work, judged on the official rubric and Core Values. Pick one, then run the FLL cycle. The roles, timeline, presentation, and rubric below apply to every project.

Team roles

Project Lead	Keeps the team on the plan and on time; makes sure every voice is heard.
Researchers (2)	Find reliable sources, take notes, and prepare the expert-interview questions.
Prototype Builders (2)	Gather materials and build the model; log what works and what breaks.
Presenters (2)	Turn the story into the 5-minute talk; make sure everyone has a speaking part.
Documenter	Keeps the Engineering Notebook: photos, sketches, decisions, and changes.
Core Values Captain	Watches for teamwork, inclusion, and fun; reminds the team of Gracious Professionalism.

Compressed 5-session timeline (time is short)

Session 1 — Identify	Pick the problem, research it, and write one clear sentence on WHAT it is and WHY it matters for biodiversity. Line up an expert to interview.
Session 2 — Design	Brainstorm 5+ ideas, choose one, and sketch a project plan. List the materials you'll need.
Session 3 — Create	Build prototype v1. Take photos. Write down every design choice in the notebook.

- Session 4 — Iterate** Interview your expert or another team, test the prototype, measure results, and make at least one improvement you can point to.
- Session 5 — Communicate** Write and rehearse the 5-minute presentation. Assign speaking parts. Practice the judges' questions.

The 5-minute presentation template

Beat	What to say
Hook — 0:30	Open with the broken connection: a quick, vivid picture (or the bioluminescent 'glow') that makes judges care.
Problem — 1:00	Name the biodiversity problem and why it matters. Show your research and one reliable source.
Solution — 1:30	Explain your innovative solution and show the prototype. Say exactly what it does and how it helps biodiversity.
Iterate — 1:00	Share the expert feedback or test result, and the improvement you made because of it.
Impact & Close — 1:00	Tie back to the vital connection you're restoring, mention a Core Value in action, and thank the judges. Everyone speaks.

Core Values & rubric

Judges score **Identify, Design, Create, Iterate, Communicate** plus **Innovation**. Show your journey, involve every kid, and keep it Gracious.

Prepared for STEM4U coaches — TurtleBricks & OrcaBricks. Starters only; the project must be the team’s original work.

All projects — index

This booklet has **34 full packets**: 10 forest, 10 urban, 10 rural (Part 1) and 4 cross-tier flagships (Part 3), plus 30 quick starter cards and the “same job, three places” map (Part 2). Every project is one broken **vital connection** and how to restore it. Projects appear in the order below.

Forest — 10

1. Pollinator Skybridges
2. Glow Guides (protect the night shift)
3. Root & Fungus Rescue
4. Seed Sowers
5. Leafcutter Lifelines
6. Keystone Watch
7. Microhabitat Minders
8. Old-Growth Guardians
9. Canopy Census
10. Living Walls

Urban — 10

1. Pollinator Alley
2. City Bug Hotels & Wild Corners
3. Balcony & Bus-Stop Blooms
4. Tree-Pit Rescue
5. Storm-Drain Guardians
6. Dark-Sky Streets
7. City Keystone Spotters
8. Vacant-Lot Revival
9. School Compost & No-Spray Grounds
10. Sidewalk BioBlitz

Rural — 10

1. Hedgerow Highway
2. Brush-Pile Havens & Nest Boxes
3. Field-Edge Wildflower Strips
4. Cover-Crop & No-Till
5. Stream Buffers & Pond Edges
6. Farmyard Dark-Sky
7. Farm Keystone Watch
8. Restore the Back Forty
9. Beneficial-Bug Farming (IPM)
10. Farm BioBlitz & Pond Dip

Cross-tier flagships — 4

1. Follow the Bee — a pollinator highway across all three
2. Follow the Raindrop — one watershed, three tiers
3. Follow the Seed — regeneration across the gradient
4. The Corridor Challenge — map & mend a real route

Part 1 — Thirty full packets · Forest, Urban, Rural

Thirty complete, judge-ready packets — the same ten jobs worked in each setting. Forest 1-10, then Urban 1-10, then Rural 1-10. (Part 2 offers the same ideas as quick starter cards; Part 3 links them across tiers.)

Forest — 10 full packets

1. Pollinator Skybridges

Problem Roads and farms cut forests into pieces, so pollinators can't cross to reach flowering trees. **Solution** Planted "canopy corridors" — flowering rope-and-vine bridges that reconnect forest fragments.

Mission M8 Tangled + M4 Lucky Leaves **Connects** BIOBUZZ ★, BIOCORE **Prototype** Model canopy bridge with flowering "stepping-stone" nodes.

1 • Identify

When a forest is split by a road, the two halves slowly become strangers. Pollinators like bees and butterflies won't fly across wide open gaps, so flowers on one side never get pollinated by insects from the other. Over time, fewer seeds form and both plants and pollinators decline — a broken connection.

RESEARCH QUESTIONS

1. How wide a gap will common pollinators refuse to cross?
2. Where have wildlife corridors or green bridges already worked?
3. Which flowering plants would make the best 'stepping stones' locally?

2 • Interview an expert

Who: A conservation-land manager, a beekeeper, or a county parks naturalist.

ASK

1. What breaks apart local habitats the most around here?
2. Have you seen wildlife or pollinator corridors work? What made them succeed?
3. Which local flowers do pollinators visit most?
4. If we built a small pollinator corridor at our school, where should it go?
5. What mistake do beginners make when trying to help pollinators?

3 • Create — prototype

MATERIALS

- Cardboard or foam board base
- String/twine for the bridge
- Green paper or fabric for vines
- Small paper/clay flowers
- Two 'forest' props (potted plants or paper trees)

BUILD STEPS

1. Set up two 'forest' patches with an open gap between them.
2. Build a bridge across the gap from string and vines.
3. Add flowering 'stepping-stone' nodes along the bridge.
4. Add paper pollinators to show the path they now can take.
5. Label the parts: fragment, corridor, node, pollinator.

4 • Iterate & 5 • Communicate

Test whether people (or a marble 'pollinator') can follow the path across without a gap wider than the node spacing; then move nodes closer/farther and see what still 'connects.'

Measure: Number of flower nodes needed so the gap is never wider than a pollinator will cross.

"Start by tearing a paper forest in half in front of the judges — 'watch what happens to the bees when we do this.'"

2. Glow Guides (protect the night shift)

Problem Light pollution confuses night pollinators and drowns out the glow of bioluminescent species. **Solution** A “dark-friendly” lighting plan: shielded, amber, motion-only lights near reserves.

Mission M7 Humongous Fungus **Connects** BIOBUZZ, BIOCORE **Prototype** Model reserve edge: shielded vs. unshielded lights + glowing markers.

1 • Identify

A lot of nature works the night shift — moths and bats pollinate, and some fungi and insects glow to send signals. Bright, always-on lights scramble those signals: moths circle bulbs instead of flowers, and faint natural glows disappear in the glare. Smarter lighting protects the night connections.

RESEARCH QUESTIONS

1. Which pollinators are active at night locally?
2. How does light color/brightness change insect behavior?
3. What lighting rules do parks or 'dark-sky' towns already use?

2 • Interview an expert

Who: A park ranger, a dark-sky advocate, or an astronomy-club member.

ASK

1. What lights are near natural areas around here, and are they needed all night?
2. Have you noticed insects or animals reacting to lights?
3. What's the difference between light that helps people and light that harms wildlife?
4. If a school wanted 'wildlife-friendly' lighting, what would you change first?
5. What's an easy win most people overlook?

3 • Create — prototype

MATERIALS

- Shoebox or foam board 'reserve edge'
- 2 small LEDs/flashlights (one shielded, one bare)
- Amber cellophane
- Glow-in-the-dark paint or stickers
- Cardboard shields

BUILD STEPS

1. Build a small reserve edge with trees and a trail.
2. Place one bare light and one shielded, amber light.
3. Add glow-painted 'fungi/insects' in the dark zone.
4. Darken the room and compare: which light hides the glow?
5. Add a motion 'switch' idea (label) so lights are off when no one's there.

4 • Iterate & 5 • Communicate

In a dark room, measure how far away the glow markers are still visible under the bare light vs. the shielded amber light; record the difference.

Measure: Distance the natural 'glow' stays visible: shielded vs. unshielded lighting.

“Turn off the room lights and reveal the glowing model — 'this is what nature is trying to show us, and this is what our streetlights hide.'”

3. Root & Fungus Rescue

Problem Digging and clearing soil breaks the underground fungal network trees use to share nutrients. **Solution** A “fungi-first” replanting kit: coat seedlings with native mycorrhizae and plant in clusters.

Mission M5 Reaching Roots + M7 Humongous Fungus **Connects** BIOCORE ★ **Prototype** Soil cross-section model showing roots linked by fungal threads.

1 • Identify

Underground, tree roots are wired together by fungal threads that trade water, nutrients, and even warning signals — the ‘wood-wide web.’ When soil is dug up or cleared, that web tears, and lone replanted seedlings struggle because they’ve lost their support network. Replanting **WITH** the fungi helps the web rebuild.

RESEARCH QUESTIONS

1. What are mycorrhizal fungi and how do they help trees?
2. Do trees planted in clusters survive better than single trees?
3. How do nurseries add fungi to seedlings?

2 • Interview an expert

Who: A soil scientist, a mycologist, or a native-plant nursery grower.

ASK

1. What happens to soil life when land is cleared?
2. How do fungi and roots actually help each other?
3. Does it matter whether we plant one tree or a cluster?
4. How could kids add the right fungi when planting?
5. What’s the most common replanting mistake?

3 • Create — prototype

MATERIALS

- Clear plastic box or jar (for the cross-section)
- Soil + real or yarn ‘roots’
- White yarn/thread for fungal threads
- Seedling props or bean sprouts
- Labels/tags

BUILD STEPS

1. Fill a clear container to show a soil cross-section.
2. Place several ‘trees’ with roots reaching down.
3. Weave white thread between roots to show the fungal web.
4. Show a lone seedling with **NO** threads next to a connected cluster.
5. Add a simple ‘planting kit’ card: fungi + cluster planting.

4 • Iterate & 5 • Communicate

Sprout beans in two cups — one planted alone, one in a cluster sharing soil — and track height/health over the sessions to see if ‘connected’ plants do better.

Measure: Height/health difference between clustered vs. lone sprouts.

“Hold up two bean sprouts: ‘one of these had friends underground, and one didn’t — guess which grew taller.’”

4. Seed Sowers

Problem Forests regrow because animals spread seeds; when those animals vanish, forests can't replant. **Solution** A kid-powered seed-ball disperser + a campaign to protect seed-spreading animals.

Mission M2 Exploding Seeds + M14 Seeds of Renewal + M1 Drone Survey **Connects** BIOCORE ★ **Prototype** Seed-ball maker + a hand/slingshot disperser.

1 • Identify

Many trees rely on animals — birds, monkeys, agoutis — to eat their fruit and drop the seeds somewhere new. Take those animals away and the forest loses its 'planting crew,' so it can't recover after damage. Helping seeds travel (and protecting the animals that move them) keeps regeneration going.

RESEARCH QUESTIONS

1. Which animals are the main seed-spreaders in forests?
2. What's inside a seed ball and why does it work?
3. Which native seeds are safe and good to plant locally?

2 • Interview an expert

Who: A reforestation nonprofit volunteer, a wildlife biologist, or a park naturalist.

ASK

1. How do forests normally grow back after damage?
2. Which animals here help spread seeds?
3. Are seed balls a real tool people use? When do they work best?
4. Which native plants should kids sow, and which should we avoid?
5. How do we make sure we help, not harm, when planting?

3 • Create — prototype

MATERIALS

- Native (or bird-seed) seeds
- Clay or soil + compost
- Water
- A scoop/mold for shaping
- Optional: a safe 'launcher' (cardboard tube)

BUILD STEPS

1. Mix clay, compost, and native seeds; roll into balls.
2. Let them dry so they hold together.
3. Build a simple, safe disperser (drop-tube or gentle launcher).
4. Test how far/accurately a ball can be placed.
5. Add a mini 'protect the seed-spreaders' poster.

4 • Iterate & 5 • Communicate

Make 10 seed balls, place half in soil and half on hard ground; over the sessions, count how many sprout in each to show where dispersal actually works.

Measure: Sprouting rate of seed balls placed well vs. poorly.

"Toss a (dry) seed ball gently to a judge: 'you just became a toucan — congratulations, you're replanting the rainforest.'"

5. Leafcutter Lifelines

Problem Pesticides and habitat loss break the ant-fungus partnership that recycles forest nutrients. **Solution** A schoolyard “no-spray zone” with a native-plant buffer to protect ant-fungus partnerships.

Mission M6 Leafcutter Frenzy + M7 Humongous Fungus **Connects** BIOCORE ★ **Prototype** Diorama of an ant highway protected by a plant buffer.

1 • Identify

Leafcutter ants are tiny farmers: they carry leaf pieces underground to grow a fungus they eat, and in doing so they recycle nutrients for the whole forest. Pesticides and cleared land wipe out the ants and their fungus, and that recycling stops. Protecting small 'no-spray' zones keeps this ancient partnership alive.

RESEARCH QUESTIONS

1. What is the leafcutter ant-fungus partnership (mutualism)?
2. How do pesticides affect insects that aren't pests?
3. What is a 'buffer zone' and how does it protect insects?

2 • Interview an expert

Who: An entomologist, an organic gardener, or a university extension agent.

ASK

1. Why do some insects matter for a healthy forest or garden?
2. What happens to helpful insects when people spray pesticides?
3. How big does a safe 'no-spray' area need to be to help?
4. What plants make the best buffer?
5. What could a school realistically do this year?

3 • Create — prototype

MATERIALS

- Shoebox diorama base
- Clay/paper ants + 'fungus garden'
- String for the ant 'highway'
- Plant props for the buffer
- Warning-sign props ('no spray')

BUILD STEPS

1. Build a diorama with an ant nest and a fungus garden.
2. Add an ant 'highway' carrying leaf bits.
3. Surround it with a native-plant buffer.
4. Show a 'sprayed' side vs. a 'protected' side.
5. Add signage and a simple school no-spray pledge.

4 • Iterate & 5 • Communicate

Survey your schoolyard for insect activity in a sprayed vs. unsprayed patch (just count visible bugs over 5 minutes each) and compare.

Measure: Insect activity count: buffer/no-spray patch vs. sprayed patch.

“Hold up a leaf with a bite out of it: 'a tiny farmer did this — and the whole forest depends on her harvest.'”

6. Keystone Watch

Problem Losing one keystone species collapses many connections — but people don't know which are keystones.

Solution Kid-friendly “keystone spotter” ID cards + a citizen-science report sheet.

Mission M13 Keystone Species + M9 Research Platform Connects BIOCORE ★, BIOBUZZ Prototype Spotter cards + a one-page report form.

1 • Identify

A keystone species is one that holds up far more of the web than its size suggests — remove it and many other species fall with it (like a beaver, a sea otter, or a bee). Most people can't name their local keystone species, so they can't help protect them. Better spotting tools turn kids into biodiversity watchers.

RESEARCH QUESTIONS

1. What makes a species a 'keystone'?
2. Which local species are keystones (and why)?
3. How does citizen-science data actually get used?

2 • Interview an expert

Who: A naturalist, a state wildlife biologist, or a nature-center educator.

ASK

1. What does 'keystone species' mean in simple terms?
2. Which keystone species live near us?
3. How do scientists use reports from ordinary people?
4. What should a kid's spotter card include to be useful?
5. What's a keystone species people often overlook?

3 • Create — prototype

MATERIALS

- Index cards for ID cards
- Printed photos/drawings of local species
- Markers
- A one-page report form
- Clipboard

BUILD STEPS

1. Pick 3–5 local keystone species.
2. Make an ID card for each (photo, key features, why it matters).
3. Design a simple report form (what, where, when, how many).
4. Test the cards with classmates — can they ID the species?
5. Plan where reports would go (a teacher, a local group, iNaturalist).

4 • Iterate & 5 • Communicate

Have 10 classmates try to identify species using your cards; measure how many they get right, then revise the confusing cards and test again.

Measure: Classmate ID accuracy before vs. after you improve the cards.

“Fan out your spotter cards to the judges: ‘in 30 seconds you’ll be able to spot a species that holds up an entire ecosystem.’”

7. Microhabitat Minders

Problem “Tidying” forests and yards removes logs and rocks — the tiny homes of insects and fungi. **Solution** Build “bug hotels” and log-and-rock refuge piles, plus signs to leave deadwood.

Mission M3 Flip the Rock + M10 Fragile Microhabitats **Connects** BIOBUZZ ★ **Prototype** A working bug-hotel model from natural materials.

1 • Identify

The 'messy' parts of nature — fallen logs, rock piles, hollow stems — are actually crowded apartment buildings for insects, salamanders, and fungi, including solitary native bees that are great pollinators. When people clear all of it away for neatness, those tiny species lose their homes. Small refuges bring them back.

RESEARCH QUESTIONS

1. Which small creatures live in deadwood and rock piles?
2. How do bug hotels help solitary native bees?
3. What materials and placement make a bug hotel actually work?

2 • Interview an expert

Who: A master gardener, a nature-center educator, or a native-bee enthusiast.

ASK

1. Why are logs and rock piles important for small wildlife?
2. What do solitary bees need to nest?
3. What makes a bug hotel succeed or fail?
4. Where should we place one at our school?
5. What do people get wrong when 'cleaning up' nature?

3 • Create — prototype

MATERIALS

- Wood scraps, hollow stems/bamboo, pinecones
- A small wooden frame or box
- Drilled wood blocks (for bees)
- String/wire
- Weatherproof label

BUILD STEPS

1. Build a frame and fill sections with different natural materials.
2. Add drilled wood blocks for solitary bees.
3. Place it in a sunny, sheltered spot.
4. Add a sign explaining why 'messy is good.'
5. Check weekly and log which sections get used.

4 • Iterate & 5 • Communicate

Set up the bug hotel (or two, in different spots) and count how many chambers show insect use over the sessions to learn the best placement/materials.

Measure: Number of occupied chambers by material/ placement.

“Hold up a chunk of 'messy' deadwood: 'most people throw this away — for a native bee, this is a five-star hotel.'”

8. Old-Growth Guardians

Problem The biggest, oldest trees support the most species — yet they're often cut first. **Solution** A “tag an elder” program: map, tag, and monitor champion trees so the community protects them.

Mission M12 Forest Elder + M11 Window to the Past **Connects** BIOCORE ★ **Prototype** Model elder tree labeled with the species it hosts.

1 • Identify

One giant old tree can be home to hundreds of species — birds, insects, mosses, fungi, mammals — far more than a young tree. But because big old trees are valuable as lumber (and seen as 'in the way'), they're often removed first, taking their whole community with them. Finding and protecting 'elders' keeps those hubs alive.

RESEARCH QUESTIONS

1. Why do older, bigger trees support more biodiversity?
2. How do people identify and protect 'champion' or heritage trees?
3. Where are the oldest trees near us?

2 • Interview an expert

Who: An arborist, a city urban-forestry office, or a park historian.

ASK

1. Why does the age and size of a tree matter for wildlife?
2. How does a town decide which trees to protect?
3. How would we measure or estimate a tree's age safely?
4. Which local trees deserve 'elder' status?
5. What threatens big old trees the most?

3 • Create — prototype

MATERIALS

- Poster or cardboard 'elder tree'
- Cut-outs of species it hosts
- Measuring tape (for real trees)
- Tags/labels
- A simple map of your area

BUILD STEPS

1. Build a model elder tree.
2. Add and label the many species that live on/around it.
3. Make a simple 'elder tag' and monitoring card.
4. Map 2–3 real candidate elder trees near school.
5. Write a short 'adopt an elder' pledge for classmates.

4 • Iterate & 5 • Communicate

Measure the trunk size of several real trees, estimate which are oldest, and count visible species/nests on a big tree vs. a small one to show the difference.

Measure: Species/nests counted on a large old tree vs. a young tree.

“Point to your model tree: ‘this one tree is home to more neighbors than our whole school — and someone wants to cut it down.’”

9. Canopy Census

Problem We can't protect biodiversity we can't measure — and the treetop canopy is hard to see. **Solution** A low-cost “canopy cam” (camera on a pole/kite) + a logging sheet to count species over time.

Mission M1 Drone Survey + M9 Research Platform **Connects** BIOCORE, BIOBUZZ **Prototype** Pole/kite camera rig model + a sample data chart.

1 • Identify

Most rainforest species live high in the canopy, where they're almost impossible to count from the ground. If scientists can't measure what's up there, they can't tell whether it's healthy or in trouble. Cheap tools — a phone on a pole, a kite camera — plus a simple counting sheet let even kids gather real biodiversity data.

RESEARCH QUESTIONS

1. Why is the canopy so important and so hard to study?
2. What cheap tools do scientists use to survey hard-to-reach places?
3. How do you count species fairly (a simple survey method)?

2 • Interview an expert

Who: A biology teacher, a drone hobbyist, or a field ecologist.

ASK

1. Why is it important to measure biodiversity, not just guess?
2. What tools help study places people can't easily reach?
3. How do scientists count species without missing or double-counting?
4. How could kids collect useful data at our school?
5. What makes data trustworthy?

3 • Create — prototype

MATERIALS

- A phone or small camera
- An extendable pole or a kite/string
- Tape/clamp to attach safely
- A printed logging sheet
- Optional: graph paper

BUILD STEPS

1. Attach a camera safely to a pole or kite.
2. Raise it to photograph a tree canopy or tall shrub.
3. Use a logging sheet to count species/pollinator visits.
4. Repeat at set times to compare.
5. Chart the results to show change over time.

4 • Iterate & 5 • Communicate

Do a 5-minute canopy count at two spots (or two times of day) and compare the tallies to show your method can detect real differences.

Measure: Species/pollinator counts across spots or times (shows the method works).

“Raise your pole-cam in the room: ‘scientists spend millions to see the treetops — we did it with a phone and a stick.’”

10. Living Walls

Problem Cities replace habitat with dead concrete and glass, cutting urban wildlife off from food. **Solution** Design a school “living wall”/green roof of native flowering plants to reconnect pollinators.

Mission M15 Biocentric Architecture + M4 Lucky Leaves **Connects** BIOBUZZ ★ **Prototype** Living-wall model with native plant “tiles.”

1 • Identify

In cities, wildlife runs into walls — literally. Concrete and glass replace the flowers and shelter that pollinators need, so bees and butterflies have nowhere to refuel as they move through town. 'Living walls' and green roofs turn dead surfaces into pollinator pit-stops, stitching the city back into nature's web.

RESEARCH QUESTIONS

1. What is a living wall / green roof and where do they exist?
2. Which native flowering plants feed local pollinators?
3. How much green space do pollinators need between stops?

2 • Interview an expert

Who: A landscape architect, a native-plant society volunteer, or a school facilities manager.

ASK

1. Why is it hard for pollinators to live in cities?
2. Do living walls really help wildlife, or just look nice?
3. Which native plants would work on our school walls?
4. What does it take to keep a living wall alive?
5. Where at our school would one help most?

3 • Create — prototype

MATERIALS

- Pegboard/cardboard 'wall' base
- Small pots or fabric pockets
- Real or paper native flowers
- Soil (if using real plants)
- Labels for each plant

BUILD STEPS

1. Build a wall base with rows of planting 'tiles.'
2. Add native flowering plants (real or model).
3. Show a bare wall vs. a living wall side by side.
4. Add paper pollinators visiting the living wall.
5. Plan a real spot and care schedule at school.

4 • Iterate & 5 • Communicate

If using real plants, track which native flowers attract the most (or any) pollinators over the sessions; if using a model, survey classmates on which design they'd support and why.

Measure: Pollinator visits (or classmate support) for living wall vs. bare wall.

“Show a photo of a blank school wall: ‘to a bee flying across town, this is a desert — here’s how we turn it into an oasis.’”

Urban — 10 full packets

Ten complete packets for the city setting.

Urban 1 — Pollinator Alley

Problem City green space is scattered in tiny patches; pollinators can't cross wide streets and parking lots to find flowers. **Solution** Create a "pollinator alley" — a chain of native planter boxes and curb/median plantings that links parks, schoolyards, and gardens into a route.

Mission M8 Tangled + M4 Lucky Leaves **Connects** BIOBUZZ ★, BIOCORE **Prototype** Block map + model with planter "stepping stones" linking two green patches.

1 • Identify

In a city, the flowers pollinators need are broken into tiny islands — a park here, a garden there — separated by streets and parking lots a small bee won't cross. Cut off, pollinators can't find enough food and city plants don't get pollinated. The connection at risk is a continuous route of blooms across the neighborhood.

RESEARCH QUESTIONS

1. How far will city bees fly between flowers?
2. Which native plants thrive in planters and medians?
3. Where are the green patches near our school, and how far apart?

2 • Interview an expert

Who: A city parks/greening staffer, a master gardener, or a beekeeper.

ASK

1. What green spaces are near our school?
2. How far can pollinators travel between them?
3. Which plants survive in city planters?
4. Where would a stepping-stone planter help most?
5. Who maintains city plantings?

3 • Create — prototype

MATERIALS

- A neighborhood map or printout
- Small boxes or cups as planters
- Paper or clay flowers
- String to mark the route
- Two 'green patch' props

BUILD STEPS

1. Map two real green patches and the gap between them.
2. Place planter 'stepping stones' along a route.
3. Add native blooms to each planter.
4. Add paper pollinators following the path.
5. Label the alley and its plants.

4 • Iterate & 5 • Communicate

On your map, measure the largest gap between blooms along the route; add planters until no gap is wider than a bee will cross, and count how many it takes.

Measure: Largest bloom-to-bloom gap on the route, before vs. after planters.

"There's a park four blocks from our school — but for a bee, it might as well be on the moon. Here's how we build the bridge."

Urban 2 — City Bug Hotels & Wild Corners

Problem Cities are swept clean of the logs, leaves, and bare soil that small creatures and solitary bees need. **Solution** Install bug hotels and leave an unmowed “wild corner” on school grounds, with signs explaining why messy helps. **Mission** M3 Flip the Rock + M10 Fragile Microhabitats **Connects** BIOBUZZ ★ **Prototype** A working bug hotel + a signed wild-corner plot.

1 • Identify

Cities are swept clean — mowed lawns, raked leaves, sealed pavement — leaving nowhere for the insects, solitary bees, and small creatures that need logs, bare soil, and hollow stems. Without those tiny homes, urban biodiversity collapses at the smallest scale. The connection at risk is shelter for the base of the food web.

RESEARCH QUESTIONS

1. Which small creatures need deadwood, bare soil, or hollow stems?
2. How do bug hotels help solitary bees?
3. What placement and materials work best?

2 • Interview an expert

Who: A master gardener, a native-bee enthusiast, or a nature-center educator.

ASK

1. Why do cities lose small wildlife?
2. What do solitary bees need to nest?
3. What makes a bug hotel work or fail?
4. Where should we place one at school?
5. What do people get wrong when 'cleaning up'?

3 • Create — prototype

MATERIALS

- Wood scraps, hollow stems/bamboo, pinecones
- A small frame or box
- Drilled wood blocks for bees
- String or wire
- A weatherproof sign

BUILD STEPS

1. Build a frame and fill sections with natural materials.
2. Add drilled blocks for solitary bees.
3. Leave an unmowed 'wild corner' nearby.
4. Place it in a sunny, sheltered spot.
5. Add a sign and log which sections get used.

4 • Iterate & 5 • Communicate

Set up the hotel (and a wild corner) and count how many chambers/areas show insect use over the sessions to learn the best placement and materials.

Measure: Number of occupied chambers by material and placement.

“Most people throw this away — but to a native bee, this piece of deadwood is a five-star hotel, and our city bulldozed all of them.”

Urban 3 — Balcony & Bus-Stop Blooms

Problem Bees and butterflies moving through the city have nowhere to refuel between parks. **Solution** Native flowering pots at bus stops, windowsills, and balconies as pollinator “fuel stations” a few blocks apart.

Mission M4 Lucky Leaves + M15 Biocentric Architecture **Connects** BIOBUZZ ★ **Prototype** A pollinator pot + a map of fuel stations.

1 • Identify

Bees and butterflies crossing a city need to refuel, but between parks there are blocks of concrete with nothing to eat. Tiny bloom 'fuel stops' on balconies, windowsills, and bus stops could keep them going — but almost no one plants for pollinators. The connection at risk is a chain of food stops across the built-up city.

RESEARCH QUESTIONS

1. Which native flowers grow well in pots and small spaces?
2. How far apart can pollinator 'fuel stops' be?
3. What blooms across the whole season?

2 • Interview an expert

Who: A native-plant nursery grower, a master gardener, or a pollinator-group volunteer.

ASK

1. Where do city pollinators struggle to find food?
2. Which flowers do best in containers?
3. How far can a butterfly go between meals?
4. Where would fuel stops help most near us?
5. What blooms early and late in the year?

3 • Create — prototype

MATERIALS

- Pots or containers
- Native flowering plants or seeds
- Potting soil
- A neighborhood map
- Plant labels

BUILD STEPS

1. Choose native blooms for containers.
2. Plant a demo pot.
3. Map candidate 'fuel stop' spots a block or two apart.
4. Add bloom-season labels.
5. Propose a set of stops that covers the whole season.

4 • Iterate & 5 • Communicate

Track which flowers actually draw pollinators (count visits over the sessions) and which spacing keeps a route 'fed'; keep the winners.

Measure: Pollinator visits per flower type (and season coverage of the route).

“A monarch flying across town is looking for a gas station — and we paved them all. Let's reopen a few.”

Urban 4 — Tree-Pit Rescue

Problem Street trees are stuck in tiny compacted pits; their roots and the soil fungi that feed them suffocate. **Solution** Mulch, de-compact, and “green” tree pits, and link them so roots and soil life can spread.

Mission M5 Reaching Roots **Connects** BIOCORE ★ **Prototype** Cross-section model of a healthy vs. compacted tree pit.

1 • Identify

Street trees are jammed into tiny squares of hard, compacted dirt where roots can't breathe and the soil fungi that feed them die off — so city trees are stressed, short-lived, and support little life. The connection at risk is the underground root-and-fungus web beneath the sidewalk.

RESEARCH QUESTIONS

1. Why do street trees struggle in compacted pits?
2. What do roots and soil fungi need?
3. How do cities improve tree pits (mulch, structural soil)?

2 • Interview an expert

Who: An arborist, an urban-forestry office, or a soil scientist.

ASK

1. Why are street trees often unhealthy?
2. What's happening underground in a tree pit?
3. How can mulch or de-compaction help?
4. Which trees near us look stressed?
5. Who cares for city trees?

3 • Create — prototype

MATERIALS

- A clear container for a cross-section
- Soil + yarn 'roots'
- White thread for fungi
- A mulch sample
- A tree prop

BUILD STEPS

1. Build a cross-section of a compacted vs. healthy tree pit.
2. Show roots choked vs. spreading.
3. Add mulch and a wider pit.
4. Show fungal threads returning.
5. Make a 'tree-pit care' card.

4 • Iterate & 5 • Communicate

Compare two potted plants — one in packed soil, one in loose mulched soil — and track growth and health over the sessions.

Measure: Growth/health of packed-soil vs. loose-soil test plants.

“This street tree is standing in a concrete coffin. Here's what's happening under the sidewalk — and how we fix it.”

Urban 5 — Storm-Drain Guardians

Problem City rain rushes off pavement carrying trash and pollutants straight into streams, harming aquatic life.

Solution Build a rain garden and run an “only rain down the drain” storm-drain marking campaign.

Mission M10 Fragile Microhabitats **Connects** BIOCORE ★ **Prototype** A model rain garden + a storm-drain marker.

1 • Identify

When it rains in the city, water sheets off roofs and pavement straight into storm drains — carrying trash, oil, and fertilizer — and dumps it untreated into local streams, harming everything that lives there. The connection at risk is clean water linking the city to its creeks.

RESEARCH QUESTIONS

1. Where does our storm-drain water go?
2. What pollutants does city runoff carry?
3. How do rain gardens clean and slow water?

2 • Interview an expert

Who: A watershed/creek group volunteer, a stormwater/public-works staffer, or a hydrologist.

ASK

1. Where does rain from our parking lot end up?
2. What does it pick up on the way?
3. How does a rain garden help?
4. Which drains near us feed a stream?
5. How do we know the water's cleaner?

3 • Create — prototype

MATERIALS

- A tray or board with a slope
- Soil + food coloring as 'pollution'
- Sponge/plant props
- A cup to catch runoff
- A storm-drain marker prop

BUILD STEPS

1. Build a sloped 'city block' model running to a drain.
2. Sprinkle 'pollution' across it.
3. Pour rain and catch the dirty runoff.
4. Add a rain garden between block and drain.
5. Pour again and compare clarity.

4 • Iterate & 5 • Communicate

Compare runoff clarity caught at the drain before vs. after adding a rain garden, and measure the difference.

Measure: Runoff clarity/volume at the drain, before vs. after a rain garden.

“Only rain should go down this drain — watch what actually does, and where it ends up.”

Urban 6 — Dark-Sky Streets

Problem Bright city lights disorient migrating birds and night pollinators and erase the stars. **Solution** Shielded, warm, motion-only lighting plus “lights-out” nights during bird migration.

Mission M7 Humongous Fungus **Connects** BIOBUZZ, BIOCORE **Prototype** Model street: shielded vs. glary lights.

1 • Identify

Bright, unshielded city lights blaze all night, disorienting migrating birds (which crash into lit buildings) and night pollinators like moths, and erasing the stars. Smarter lighting protects wildlife and still keeps people safe. The connection at risk is the natural darkness that night wildlife depends on.

RESEARCH QUESTIONS

1. How does light pollution affect birds and night pollinators?
2. What lighting is 'wildlife-friendly'?
3. What do dark-sky towns or buildings already do?

2 • Interview an expert

Who: A dark-sky advocate, an astronomer, or a city planner.

ASK

1. What lights near us stay on all night?
2. How do lights affect birds and insects?
3. What's the difference between helpful and harmful lighting?
4. What could our school change first?
5. When is bird migration here?

3 • Create — prototype

MATERIALS

- A model street or building
- Two lights (bare + shielded)
- Amber cellophane
- Bird/moth props
- A 'lights-out' sign

BUILD STEPS

1. Build a model street with a lit building.
2. Show birds/moths drawn to a bare light.
3. Swap in shielded, amber, motion lighting.
4. Show fewer collisions.
5. Add a 'lights-out during migration' pledge.

4 • Iterate & 5 • Communicate

In a dark room, compare how far a bare vs. shielded light throws glare and how many 'moth' tokens it draws; record the difference.

Measure: Glare distance / 'moths' drawn: bare vs. shielded light.

“Every spring, birds fly into our brightest buildings by the thousands. It takes one switch to stop it.”

Urban 7 — City Keystone Spotters

Problem People don't realize cities have keystone species (street trees, urban bees, peregrine falcons) worth protecting. **Solution** Kid-made spotter cards + a report form for urban keystone species.

Mission M13 Keystone Species + M9 Research Platform **Connects** BIOCORE ★, BIOBUZZ **Prototype** Urban spotter cards + a report form.

1 • Identify

Cities do have keystone species — big street trees that cool and feed a whole block, urban bees, even peregrine falcons — but most residents can't name them, so they aren't valued or protected. The connection at risk is public awareness of the species holding the urban web together.

RESEARCH QUESTIONS

1. What is a keystone species, and which are urban?
2. Why do big street trees act like keystones?
3. How is citizen-science data used?

2 • Interview an expert

Who: A naturalist, an urban ecologist, or a nature-center educator.

ASK

1. Which species do a lot for our neighborhood?
2. Why are big trees so important in a city?
3. How do scientists use public sightings?
4. What should a spotter card show?
5. Which urban keystone is overlooked?

3 • Create — prototype

MATERIALS

- Index cards
- Photos/drawings of local species
- Markers
- A one-page report form
- A clipboard

BUILD STEPS

1. Pick 3–5 urban keystone species.
2. Make an ID card for each.
3. Design a simple report form.
4. Test the cards with classmates.
5. Plan where reports go (teacher, iNaturalist, city).

4 • Iterate & 5 • Communicate

Have 10 classmates ID species using your cards; measure accuracy, fix the confusing cards, and test again.

Measure: Classmate ID accuracy, before vs. after improving the cards.

"This one tree cools our whole block, feeds birds, and cleans our air — and nobody knows its name. Let's change that."

Urban 8 — Vacant-Lot Revival

Problem Empty city lots sit dead, weedy, or paved — lost habitat in the heart of the neighborhood. **Solution** Turn a vacant lot (or unused school corner) into a native pocket meadow with seed balls and community help.

Mission M14 Seeds of Renewal + M2 Exploding Seeds **Connects** BIOCORE ★, BIOBUZZ **Prototype** Before/after model + a seed-ball plan for a real lot.

1 • Identify

Empty city lots sit paved, weedy, or trash-strewn — dead zones in the middle of a neighborhood that could be buzzing habitat. Turning even one into a native pocket meadow reconnects urban wildlife and gives people nature nearby. The connection at risk is habitat in the heart of the city.

RESEARCH QUESTIONS

1. What native plants make a good pocket meadow?
2. How do seed balls establish plants cheaply?
3. Who owns or permits vacant lots?

2 • Interview an expert

Who: A community-garden or land-trust volunteer, a restoration ecologist, or a city official.

ASK

1. What could an empty lot become?
2. Which native plants would thrive there?
3. How do we start plants cheaply?
4. Who do we ask for permission?
5. What keeps a meadow healthy?

3 • Create — prototype

MATERIALS

- Native seeds
- Clay + compost for seed balls
- A photo/map of a real lot
- Before/after sketch materials
- A plan sheet

BUILD STEPS

1. Pick a real vacant lot or unused school corner.
2. Make native seed balls.
3. Sketch a before/after design.
4. Plan planting and who helps.
5. Add a 'pocket meadow' sign.

4 • Iterate & 5 • Communicate

Sow seed balls in a test patch vs. bare ground; over the sessions, count sprouts to show the method works before proposing the full lot.

Measure: Seed-ball sprouting rate in the test patch.

"Everyone walks past this empty lot. In one season, we could make it hum with bees."

Urban 9 — School Compost & No-Spray Grounds

Problem Schools haul away leaves and food scraps and spray their grounds, starving decomposers and harming pollinators. **Solution** Start a compost system and a pesticide-free grounds pledge so nutrients cycle and helpful bugs survive.

Mission M6 Leafcutter Frenzy + M7 Humongous Fungus **Connects** BIOCORE ★, BIOBUZZ **Prototype** A working mini-compost bin + a no-spray zone sign.

1 • Identify

Schools haul away leaves and food scraps and spray their lawns — starving the decomposers that recycle nutrients and poisoning the pollinators that visit. Composting and going spray-free keeps that cycle alive right on campus. The connection at risk is the nutrient-and-pollinator cycle in our own grounds.

RESEARCH QUESTIONS

1. How does composting recycle nutrients?
2. How do pesticides harm non-pest insects?
3. What does a pesticide-free grounds plan involve?

2 • Interview an expert

Who: A master gardener, a school facilities manager, or an extension agent.

ASK

1. What happens to our leaves and food scraps now?
2. How does compost help the soil and bugs?
3. What does spraying do to bees and worms?
4. Could our school go spray-free?
5. What's an easy first step?

3 • Create — prototype

MATERIALS

- A small compost bin or model
- 'Green'/'brown' material samples
- A no-spray zone sign
- A simple pledge sheet

BUILD STEPS

1. Set up a small compost bin.
2. Sort greens and browns.
3. Track it breaking down.
4. Mark a 'no-spray' zone.
5. Write a grounds pledge for the school.

4 • Iterate & 5 • Communicate

Compare decomposition in a compost bin vs. a sealed bag over the sessions (or worm activity in composted vs. bare soil) and record what you see.

Measure: Decomposition/worm activity: composted vs. control.

"Our cafeteria throws away lunch scraps and our lawn gets sprayed — meanwhile the soil is starving. Here's the fix, on our own grounds."

Urban 10 — Sidewalk BioBlitz

Problem We can't protect city nature we haven't counted — and most people think cities have none. **Solution** Run a neighborhood "BioBlitz," logging species with a phone app (iNaturalist) over time.

Mission M1 Drone Survey + M9 Research Platform **Connects** BIOCORE, BIOBUZZ **Prototype** A BioBlitz logging sheet + a species tally chart.

1 • Identify

People assume cities have no nature, so no one measures it — and you can't protect what you never counted. A neighborhood 'BioBlitz' reveals the surprising biodiversity on our own sidewalks and gives real data to act on. The connection at risk is knowing (and valuing) the life already around us.

RESEARCH QUESTIONS

1. What is a BioBlitz and how is it run?
2. What tools (like iNaturalist) log species?
3. How do you count species fairly?

2 • Interview an expert

Who: A biology teacher, a naturalist, or a citizen-science organizer.

ASK

1. How much wildlife is really around our school?
2. What tools help identify species?
3. How do we count without double-counting?
4. How is our data used?
5. What surprises people most?

3 • Create — prototype

MATERIALS

- A phone or camera
- A printed logging sheet
- A marked survey route
- Graph paper for a tally

BUILD STEPS

1. Pick a survey route around school.
2. Log every species with photos/app.
3. Repeat at set times.
4. Tally by group (birds, bugs, plants).
5. Chart the results.

4 • Iterate & 5 • Communicate

Compare species counts at two spots or two times of day to show your method detects real differences.

Measure: Species counts across spots/times (shows the method works).

"People say our neighborhood has no nature. Give us 30 minutes and a phone — we'll prove them wrong."

Rural — 10 full packets

Ten complete packets for the countryside setting.

Rural 1 — Hedgerow Highway

Problem Big open fields leave wildlife with no cover to travel between woodlots, so populations get cut off. **Solution** Plant or restore hedgerows and field-edge shrubs as wildlife corridors linking habitat patches.

Mission M8 Tangled + M4 Lucky Leaves Connects BIOBUZZ ★, BIOCORE Prototype Farm map + model hedgerow linking two woodlots.

1 • Identify

Modern farms are wide-open fields with no cover, so wildlife trying to move between woodlots has nowhere to hide and won't cross — trapping populations on shrinking islands. Hedgerows and shrubby field edges restore the corridors that let animals and pollinators travel. The connection at risk is safe passage across farmland.

RESEARCH QUESTIONS

1. What is a hedgerow and how does it help wildlife?
2. Which native shrubs make good hedgerows here?
3. How far can wildlife safely travel with cover?

2 • Interview an expert

Who: A conservation-district staffer, a farmer, or a wildlife biologist.

ASK

1. What stops wildlife from crossing open fields?
2. Where are the woodlots that should be linked?
3. Which shrubs work as a hedgerow?
4. How wide does it need to be?
5. What else does a hedgerow do for a farm?

3 • Create — prototype

MATERIALS

- A farm map or sketch
- Model shrubs/hedgerow
- Two 'woodlot' props
- Animal/pollinator tokens

BUILD STEPS

1. Map two woodlots separated by open field.
2. Add a hedgerow linking them.
3. Place shrubs and flowers along it.
4. Move animal/pollinator tokens along the safe route.
5. Label the highway.

4 • Iterate & 5 • Communicate

On the map, measure the open distance wildlife must cross before vs. after the hedgerow; show the route now has continuous cover.

Measure: Longest open (no-cover) stretch on the route, before vs. after.

"A rabbit at the edge of this field can see the woods across the way — and no safe way to get there. Let's build the road."

Rural 2 — Brush-Pile Havens & Nest Boxes

Problem “Clean” farms remove brush, dead snags, and old barns — the shelter small animals, owls, and bats need.

Solution Build brush piles and put up barn-owl and bat boxes to bring back farm helpers.

Mission M3 Flip the Rock + M10 Fragile Microhabitats **Connects** BIOCORE ★, BIOBUZZ **Prototype** A model brush pile + a nest/bat box.

1 • Identify

'Tidy' farms clear the brush, dead snags, and old outbuildings that shelter rabbits, songbirds, owls, and bats — losing the very helpers that control pests. Brush piles and nest/bat boxes bring them back. The connection at risk is farm shelter for beneficial wildlife.

RESEARCH QUESTIONS

1. Which farm animals rely on brush, snags, or old barns?
2. How do barn owls and bats help farms?
3. What makes a good brush pile or nest box?

2 • Interview an expert

Who: A wildlife biologist, an extension agent, or a farmer who uses owl/bat boxes.

ASK

1. What shelter do farm animals lose when we tidy up?
2. How do owls and bats help control pests?
3. What makes a nest or bat box work?
4. Where should we place them?
5. What's the payoff for the farmer?

3 • Create — prototype

MATERIALS

- Branches for a model brush pile
- A model nest/bat box
- A farm map
- Animal props

BUILD STEPS

1. Build a model brush pile in a field corner.
2. Add a nest box and a bat box.
3. Place them where wildlife needs cover.
4. Show the pest-eating helpers they attract.
5. Make a 'leave some wild' sign.

4 • Iterate & 5 • Communicate

If possible, set a real brush pile or box and log wildlife use over the sessions; otherwise survey a 'kept-wild' corner vs. a mowed one for signs of life.

Measure: Wildlife signs at a brush pile/box (or wild corner vs. mowed).

“Farmers pay for pest control — but an owl works for free, if we just give her a place to live.”

Rural 3 — Field-Edge Wildflower Strips

Problem Crop monocultures offer almost no flowers, so pollinators — and the crops that need them — suffer. **Solution** Plant native wildflower strips around and between fields as pollinator refuel and habitat.

Mission M4 Lucky Leaves + M2 Exploding Seeds **Connects** BIOBUZZ ★ **Prototype** A field diagram with wildflower borders + a seed-mix plan.

1 • Identify

A field of a single crop is a food desert for pollinators — nothing blooms for most of the year — so bees decline and even the crops that need pollination suffer. Native wildflower strips along field edges feed pollinators and beneficial insects. The connection at risk is pollinator food on farmland.

RESEARCH QUESTIONS

1. Why are crop fields poor for pollinators?
2. Which native wildflowers feed them across the season?
3. How wide should a flower strip be?

2 • Interview an expert

Who: An extension agent, a pollinator/conservation group, or a farmer.

ASK

1. Why do pollinators struggle around crops?
2. Which flowers help most, and when?
3. How wide a strip makes a difference?
4. Where on a field should strips go?
5. How does this help the crop?

3 • Create — prototype

MATERIALS

- A field diagram
- Native wildflower seed samples
- Model flower strips
- Pollinator tokens

BUILD STEPS

1. Diagram a crop field.
2. Add wildflower strips along the edges and a center row.
3. Label the bloom seasons.
4. Add pollinators visiting.
5. Propose a seed mix.

4 • Iterate & 5 • Communicate

Compare pollinator visits at a flowering edge vs. a bare edge (real, or modeled with a bloom-time chart) to show the strips fill the gaps.

Measure: Pollinator visits (or bloom-weeks): flowering edge vs. bare edge.

"This whole field feeds people for two weeks a year. For a bee, that's a famine. Here's the fix that also grows more crops."

Rural 4 — Cover-Crop & No-Till

Problem Bare, tilled soil between crops breaks the underground fungal web and washes away. **Solution** Plant cover crops and reduce tilling so the soil's living web (roots + fungi) stays intact.

Mission M5 Reaching Roots + M7 Humongous Fungus **Connects** BIOCORE ★ **Prototype** Two soil jars — tilled/bare vs. cover-cropped — showing roots + structure.

1 • Identify

Between harvests, bare tilled soil bakes, blows away, and loses the living fungal web that holds it together and feeds the next crop. Cover crops and less tilling keep the soil alive. The connection at risk is the underground soil web on farmland.

RESEARCH QUESTIONS

1. What do cover crops do for soil?
2. How does tilling harm soil fungi and structure?
3. Which cover crops work locally?

2 • Interview an expert

Who: A soil scientist, an extension agent, or a no-till farmer.

ASK

1. What happens to bare soil between crops?
2. How does tilling affect the soil's living web?
3. What do cover crops add?
4. Which ones grow well here?
5. What convinces a farmer to switch?

3 • Create — prototype

MATERIALS

- Two clear jars
- Soil
- Grass/seed for a cover crop
- Water
- Labels

BUILD STEPS

1. Fill two jars: one bare/tilled, one with a growing cover crop.
2. Compare roots and structure.
3. Pour water on each to test erosion.
4. Show which holds together.
5. Make a 'keep it covered' card.

4 • Iterate & 5 • Communicate

Pour equal water over bare vs. cover-cropped soil trays and measure the runoff and soil lost from each.

Measure: Runoff/soil lost: bare vs. cover-cropped soil.

"Between crops, this field is just dirt, blowing away. Here's how farmers keep it alive underground."

Rural 5 — Stream Buffers & Pond Edges

Problem Farm runoff (soil, fertilizer) pours into creeks and ponds, choking aquatic life. **Solution** Plant “buffer” strips of native grasses and trees along waterways to filter runoff.

Mission M10 Fragile Microhabitats + M5 Reaching Roots **Connects** BIOCORE ★ **Prototype** A model creek with vs. without a plant buffer (pour muddy water to test).

1 • Identify

Rain washes soil and fertilizer off farm fields straight into creeks and ponds, clouding the water and choking fish and frogs. A 'buffer' strip of native plants along the water filters and slows the runoff. The connection at risk is clean farm water.

RESEARCH QUESTIONS

1. How does farm runoff harm creeks and ponds?
2. How do buffer strips filter water?
3. How wide should a buffer be?

2 • Interview an expert

Who: A soil & water conservation district staffer, a watershed group, or a farmer.

ASK

1. What runs off fields into the water?
2. How do plant buffers help?
3. How wide is wide enough?
4. Which streams/ponds near us need it?
5. How do we tell if it's working?

3 • Create — prototype

MATERIALS

- A tilted tray 'field to creek'
- Soil + food coloring
- Grass/plant props for a buffer
- A cup to catch water

BUILD STEPS

1. Build a sloped field-to-creek model.
2. Sprinkle 'soil and fertilizer'.
3. Pour rain and catch the dirty runoff.
4. Add a plant buffer strip.
5. Pour again and compare.

4 • Iterate & 5 • Communicate

Compare the clarity and soil in runoff caught at the 'creek' with vs. without the buffer.

Measure: Runoff clarity/soil at the creek, with vs. without a buffer.

"After a storm, this creek turns brown with the farm's topsoil. A strip of grass could keep it — and the fish — in place."

Rural 6 — Farmyard Dark-Sky

Problem All-night barn and yard lights wipe out fireflies, moths, and the bats that eat crop pests. **Solution** Shield and motion-switch farm lights to protect night wildlife — and get free pest control.

Mission M7 Humongous Fungus **Connects** BIOBUZZ, BIOCORE **Prototype** Model farmyard: glary vs. shielded lights + a firefly-count plan.

1 • Identify

Barn and yard lights left on all night wipe out fireflies, moths, and the bats that eat crop pests — trading a little convenience for free pest control and a starry sky. Shielded, motion-triggered lights fix it. The connection at risk is the farm's night wildlife.

RESEARCH QUESTIONS

1. How does night lighting affect fireflies, moths, and bats?
2. How do bats help farms?
3. What lighting is wildlife-friendly?

2 • Interview an expert

Who: A dark-sky advocate, a wildlife biologist, or an extension agent.

ASK

1. What farm lights stay on all night?
2. What happens to night insects and bats?
3. How do bats help with pests?
4. What lighting change is easy?
5. When are fireflies and bats most active?

3 • Create — prototype

MATERIALS

- A model farmyard
- Two lights (bare + shielded)
- Amber film
- Firefly/moth/bat props

BUILD STEPS

1. Build a model farmyard with a bright yard light.
2. Show fewer fireflies/ moths near it.
3. Swap in shielded, amber, motion lighting.
4. Show night wildlife return.
5. Add a firefly-count plan.

4 • Iterate & 5 • Communicate

In a dark room, compare how far the bare vs. shielded light glares and how many 'insect' tokens it pulls off course.

Measure: Glare distance / insects drawn: bare vs. shielded light.

"Grandpa remembers fields full of fireflies. The barn light that never turns off is why they're gone — and it's an easy fix."

Rural 7 — Farm Keystone Watch

Problem Farmland keystones — beavers, bats, dung beetles, barn owls — go unseen and are often removed. **Solution** Spotter cards + a watch log for the keystone helpers on a farm.

Mission M13 Keystone Species + M9 Research Platform **Connects** BIOCORE ★ **Prototype** Farm keystone spotter cards + a report form.

1 • Identify

Farmland keystones — beavers that hold water, bats and barn owls that eat pests, dung beetles that recycle manure — do huge unseen work, yet are often removed as nuisances. Spotting and valuing them keeps the farm web intact. The connection at risk is awareness of the farm's keystone helpers.

RESEARCH QUESTIONS

1. Which farmland species are keystones?
2. What jobs do beavers, bats, owls, and dung beetles do?
3. How is citizen-science data used?

2 • Interview an expert

Who: A wildlife biologist, an extension agent, or a naturalist.

ASK

1. Which wild species help this farm the most?
2. What does each one actually do?
3. Why are they sometimes removed?
4. How would we track them?
5. Which is most overlooked?

3 • Create — prototype

MATERIALS

- Index cards
- Photos of farm keystones
- Markers
- A report form
- A clipboard

BUILD STEPS

1. Pick 3–5 farm keystone species.
2. Make ID cards with each one's 'job'.
3. Design a watch log.
4. Test the cards with classmates.
5. Plan where reports go.

4 • Iterate & 5 • Communicate

Have classmates ID the species with your cards; measure accuracy, fix the unclear ones, and re-test.

Measure: Classmate ID accuracy, before vs. after improving cards.

"A single family of bats over this barn eats thousands of crop pests a night — and the farmer's about to seal them out. Meet the farm's hidden workforce."

Rural 8 — Restore the Back Forty

Problem Marginal or worn-out field corners sit bare or invasive-choked, lost to biodiversity. **Solution** Replant a field corner or marginal strip with native seed to rebuild habitat.

Mission M14 Seeds of Renewal + M2 Exploding Seeds **Connects** BIOCORE ★, BIOBUZZ **Prototype** A restoration plan + seed balls for a real corner.

1 • Identify

Every farm has marginal ground — a wet corner, a worn-out strip, a weedy edge — that grows little and could instead be rich native habitat. Replanting it with native seed rebuilds biodiversity without touching productive land. The connection at risk is habitat on unused farmland.

RESEARCH QUESTIONS

1. What native plants restore worn or marginal land?
2. How do seed balls / native seed establish?
3. Which areas are best to restore?

2 • Interview an expert

Who: A restoration ecologist, a land-trust volunteer, or an extension agent.

ASK

1. Which parts of a farm grow the least?
2. What native plants would thrive there?
3. How do we start them cheaply?
4. What wildlife would return?
5. What's the benefit to the farmer?

3 • Create — prototype

MATERIALS

- A farm map
- Native seeds + seed-ball materials
- Before/after sketch materials
- A plan sheet

BUILD STEPS

1. Identify a marginal corner.
2. Make native seed balls.
3. Sketch a before/after.
4. Plan planting and care.
5. Add a 'restored habitat' marker.

4 • Iterate & 5 • Communicate

Sow native seed balls in a test patch vs. bare ground; count sprouts over the sessions before proposing the full corner.

Measure: Seed-ball sprouting rate in the test patch.

"Nothing grows in the back corner of this field — which makes it the perfect place to grow a meadow."

Rural 9 — Beneficial-Bug Farming (IPM)

Problem Broad pesticide spraying kills the good bugs — ladybugs, bees, decomposers — along with the pests.

Solution Use Integrated Pest Management: invite beneficial insects and spot-treat instead of blanket spraying.

Mission M6 Leafcutter Frenzy + M7 Humongous Fungus **Connects** BIOCORE ★, BIOBUZZ **Prototype** A “good bug vs. pest” guide + an insectary-plant plan.

1 • Identify

Blanket pesticide spraying kills the good bugs — ladybugs, lacewings, bees, decomposers — along with the pests, so the farm loses its free workforce and its pollinators. Integrated Pest Management invites beneficial insects and sprays only as a last resort. The connection at risk is the balance of helpful and harmful insects.

RESEARCH QUESTIONS

1. What is Integrated Pest Management (IPM)?
2. Which 'good bugs' control pests?
3. What plants attract beneficial insects?

2 • Interview an expert

Who: An entomologist, an extension agent, or an IPM farmer.

ASK

1. What does broad spraying do to helpful insects?
2. Which good bugs eat pests?
3. How do we attract them?
4. When is spraying truly needed?
5. What's the payoff of IPM?

3 • Create — prototype

MATERIALS

- A 'good bug vs. pest' guide
- Insectary-plant samples
- A field diagram
- Bug props

BUILD STEPS

1. Make a guide sorting good bugs from pests.
2. Add insectary plants that attract beneficials.
3. Diagram where to plant them.
4. Show pests controlled without spraying.
5. Add an IPM tip card.

4 • Iterate & 5 • Communicate

Survey a sprayed vs. unsprayed patch (or a flowering vs. bare edge) for beneficial insects over 5 minutes each and compare.

Measure: Beneficial-insect count: IPM/flowering patch vs. sprayed patch.

“One spray to kill the pests — and it takes out the ladybugs, the bees, and the farmer's best free help. Here's the smarter way.”

Rural 10 — Farm BioBlitz & Pond Dip

Problem Farmers rarely measure the wild biodiversity on their land, so they can't tell if it's thriving. **Solution** Run a farm BioBlitz — a pond dip, a night moth sheet, a bird count — logged over time.

Mission M1 Drone Survey + M9 Research Platform **Connects** BIOCORE, BIOBUZZ **Prototype** A pond-dip kit + a species tally chart.

1 • Identify

Farmers rarely measure the wild biodiversity on their land, so they can't tell whether it's thriving or fading. A farm BioBlitz — a pond dip, a night moth sheet, a bird count — turns the whole farm into a living dataset. The connection at risk is knowing the health of the farm's wild web.

RESEARCH QUESTIONS

1. What is a BioBlitz, and how do you run a pond dip?
2. What tools log species?
3. How do you count fairly?

2 • Interview an expert

Who: A biology teacher, a naturalist, or a watershed/citizen-science group.

ASK

1. How much wildlife lives on this farm?
2. What tools help count it?
3. How do we survey a pond?
4. How is the data used?
5. What would surprise the farmer?

3 • Create — prototype

MATERIALS

- A net + tray for a pond dip
- A white sheet + light for moths
- A logging sheet
- Graph paper

BUILD STEPS

1. Pick spots: pond, field, hedgerow, night yard.
2. Do a pond dip and a moth-sheet count.
3. Log every species.
4. Tally by group.
5. Chart the results over time.

4 • Iterate & 5 • Communicate

Compare species counts at two habitats (pond vs. field) or two times to show the survey detects real differences.

Measure: Species counts across habitats/times (shows the method works).

"This farm looks like just corn and cows — until you dip a net in the pond. Watch what comes up."

Part 2 — Three settings, ten starters each

A canopy isn't only a rainforest. There's a **forest canopy**, an **urban canopy** (street trees, parks, green roofs, vacant lots, storm drains, balconies) and a **rural canopy** (hedgerows, woodlots, field edges, ponds, pastures, ditches). The same ten ecological **jobs** appear in all three — so any team can do the season's work where they live. Read the map across a row: the identical job, done three ways.

Same job, three places

#	The job	Forest	Urban	Rural
1	Connection / corridor	Pollinator Skybridges	Pollinator Alley	Hedgerow Highway
2	Refuge / microhabitat	Microhabitat Minders	City Bug Hotels & Wild Corners	Brush-Pile Havens & Nest Boxes
3	Pollinator refuel	Light-Gap Blooms	Balcony & Bus-Stop Blooms	Field-Edge Wildflower Strips
4	Soil / root web	Root & Fungus Rescue	Tree-Pit Rescue	Cover-Crop & No-Till
5	Water web	Headwater Guardians	Storm-Drain Guardians	Stream Buffers & Pond Edges
6	Night sky / light	Glow Guides	Dark-Sky Streets	Farmyard Dark-Sky
7	Keystone watch	Keystone Watch	City Keystone Spotters	Farm Keystone Watch
8	Seed / regeneration	Seed Sowers	Vacant-Lot Revival	Restore the Back Forty
9	Nutrient / no-spray	Leafcutter Lifelines	Compost & No-Spray Grounds	Beneficial-Bug Farming (IPM)
10	Monitoring / census	Canopy Census	Sidewalk BioBlitz	Farm BioBlitz & Pond Dip
11	Heritage / old-growth	Old-Growth Guardians	Heritage street trees	Lone pasture oak / old woodlot

Row 11 (heritage / old-growth) is the forest's standout; its city and country cousins are protecting heritage street trees and lone pasture oaks.

Forest Canopy — 10 starters

1

Pollinator Skybridges

PROBLEM Roads and logging split forests into islands; pollinators and small animals can't cross the gaps.

SOLUTION Flowering canopy corridors — rope-and-vine bridges with blooms — that reconnect forest fragments.

MISSION M8 Tangled + M4 Lucky Leaves

CONNECTS BIOBUZZ ★, BIOCORE

PROTOTYPE Model canopy bridge with flowering nodes between two patches.

WHERE Fragmented forest, logging edges, canopy gaps

2

Microhabitat Minders

PROBLEM Clearing deadwood and rocks removes the shelters tiny forest species and solitary bees need.

SOLUTION Protect and build deadwood & rock refuges (“leave the logs”) and bug hotels on the forest floor.

MISSION M3 Flip the Rock + M10 Fragile Microhabitats

CONNECTS BIOBUZZ ★, BIOCORE

PROTOTYPE A deadwood/rock refuge model + a “leave it wild” sign.

WHERE Forest floor, woodland edges

3

Light-Gap Blooms

PROBLEM Deep shade under a closed canopy has few flowers, so forest pollinators crowd into rare sunny clearings that are shrinking.

SOLUTION Plant and protect flowering understory in natural light gaps and along forest edges as pollinator “refuel” spots.

MISSION M4 Lucky Leaves + M2 Exploding Seeds

CONNECTS BIOBUZZ ★

PROTOTYPE A model forest with a sunlit bloom-filled gap + a map of gap “fuel stops.”

WHERE Forest clearings, treefall gaps, woodland edges

4

Root & Fungus Rescue

PROBLEM Disturbing forest soil tears the fungal “wood-wide web” that trees use to share water and nutrients.

SOLUTION A “fungi-first” replanting kit — native mycorrhizae + cluster planting so the network rebuilds.

MISSION M5 Reaching Roots + M7 Humongous Fungus

CONNECTS BIOCORE ★

PROTOTYPE Soil cross-section model showing roots linked by fungal threads.

WHERE Replanting sites, woodland soil

5

Headwater Guardians

PROBLEM Forest streams and springs — the source of clean water for everything downstream — are damaged by erosion and disturbance.

SOLUTION Protect headwater streams and wetlands with native buffer plantings and “no-disturbance” zones.

MISSION M10 Fragile Microhabitats + M5 Reaching Roots

CONNECTS BIOCORE ★

PROTOTYPE A model forest stream with vs. without a plant buffer (pour muddy water to test).

WHERE Forest streams, springs, wetlands

6

Glow Guides

PROBLEM Light and disturbance near forests disrupt nocturnal pollinators and bioluminescent species.

SOLUTION Dark-friendly practices near forest reserves — shielded, amber, motion-only lighting.

MISSION M7 Humongous Fungus

CONNECTS BIOBUZZ, BIOCORE

PROTOTYPE Model reserve edge: shielded vs. glary light + glow markers.

WHERE Forest reserves, trailheads

7

Keystone Watch

PROBLEM Losing one forest keystone species collapses many connections, and people can't name them.

SOLUTION Keystone spotter cards + a citizen-science report sheet for forest keystone species.

MISSION M13 Keystone Species + M9 Research Platform

CONNECTS BIOCORE ★, BIOBUZZ

PROTOTYPE Forest keystone spotter cards + a report form.

WHERE Any forest or park

8

Seed Sowers

PROBLEM When seed-spreading forest animals disappear, the forest can't replant itself.

SOLUTION Seed-ball dispersal + a campaign to protect the animals that spread forest seeds.

MISSION M2 Exploding Seeds + M14 Seeds of Renewal

CONNECTS BIOCORE ★, BIOBUZZ

PROTOTYPE A seed-ball maker + disperser and a sprout test.

WHERE Damaged forest, clearings

9

Leafcutter Lifelines

PROBLEM Pesticides and clearing break the ant-fungus partnership that recycles forest nutrients.

SOLUTION Protect “no-spray” forest-floor zones with native-plant buffers to keep the partnership alive.

MISSION M6 Leafcutter Frenzy + M7 Humongous Fungus

CONNECTS BIOCORE ★, BIOBUZZ

PROTOTYPE A diorama of an ant highway protected by a plant buffer.

WHERE Forest floor, forest-farm edge

10

Canopy Census

PROBLEM Most forest species live in the canopy, which is nearly impossible to count from the ground.

SOLUTION A low-cost canopy cam (pole/kite) + a logging sheet to count species and pollinator visits over time.

MISSION M1 Drone Survey + M9 Research Platform

CONNECTS BIOCORE, BIOBUZZ

PROTOTYPE A pole/kite camera rig model + a species tally chart.

WHERE Forest canopy

Urban Canopy — 10 starters

1

Pollinator Alley

PROBLEM City green space is scattered in tiny patches; pollinators can't cross wide streets and parking lots to find flowers.

SOLUTION Create a “pollinator alley” — a chain of native planter boxes and curb/median plantings that links parks, schoolyards, and gardens into a route.

MISSION M8 Tangled + M4 Lucky Leaves

CONNECTS BIOBUZZ ★, BIOCORE

PROTOTYPE Block map + model with planter “stepping stones” linking two green patches.

WHERE School courtyard, medians, sidewalks, community garden

2

City Bug Hotels & Wild Corners

PROBLEM Cities are swept clean of the logs, leaves, and bare soil that small creatures and solitary bees need.

SOLUTION Install bug hotels and leave an unmowed “wild corner” on school grounds, with signs explaining why messy helps.

MISSION M3 Flip the Rock + M10 Fragile Microhabitats

CONNECTS BIOBUZZ ★

PROTOTYPE A working bug hotel + a signed wild-corner plot.

WHERE Schoolyard edge, park, courtyard

3

Balcony & Bus-Stop Blooms

PROBLEM Bees and butterflies moving through the city have nowhere to refuel between parks.

SOLUTION Native flowering pots at bus stops, windowsills, and balconies as pollinator “fuel stations” a few blocks apart.

MISSION M4 Lucky Leaves + M15 Biocentric Architecture

CONNECTS BIOBUZZ ★

PROTOTYPE A pollinator pot + a map of fuel stations.

WHERE Balconies, bus stops, classroom windows

4

Tree-Pit Rescue

PROBLEM Street trees are stuck in tiny compacted pits; their roots and the soil fungi that feed them suffocate.

SOLUTION Mulch, de-compact, and “green” tree pits, and link them so roots and soil life can spread.

MISSION M5 Reaching Roots

CONNECTS BIOCORE ★

PROTOTYPE Cross-section model of a healthy vs. compacted tree pit.

WHERE Street trees near school

5

Storm-Drain Guardians

PROBLEM City rain rushes off pavement carrying trash and pollutants straight into streams, harming aquatic life.

SOLUTION Build a rain garden and run an “only rain down the drain” storm-drain marking campaign.

MISSION M10 Fragile Microhabitats

CONNECTS BIOCORE ★

PROTOTYPE A model rain garden + a storm-drain marker.

WHERE School downspouts, parking lots, curbs

6

Dark-Sky Streets

PROBLEM Bright city lights disorient migrating birds and night pollinators and erase the stars.

SOLUTION Shielded, warm, motion-only lighting plus “lights-out” nights during bird migration.

MISSION M7 Humongous Fungus

CONNECTS BIOBUZZ, BIOCORE

PROTOTYPE Model street: shielded vs. glary lights.

WHERE School exterior lights, parking lots

7

City Keystone Spotters

PROBLEM People don't realize cities have keystone species (street trees, urban bees, peregrine falcons) worth protecting.

SOLUTION Kid-made spotter cards + a report form for urban keystone species.

MISSION M13 Keystone Species + M9 Research Platform

CONNECTS BIOCORE ★, BIOBUZZ

PROTOTYPE Urban spotter cards + a report form.

WHERE Neighborhood, park, school grounds

8

Vacant-Lot Revival

PROBLEM Empty city lots sit dead, weedy, or paved — lost habitat in the heart of the neighborhood.

SOLUTION Turn a vacant lot (or unused school corner) into a native pocket meadow with seed balls and community help.

MISSION M14 Seeds of Renewal + M2 Exploding Seeds

CONNECTS BIOCORE ★, BIOBUZZ

PROTOTYPE Before/after model + a seed-ball plan for a real lot.

WHERE Vacant lot, unused school ground

9

School Compost & No-Spray Grounds

PROBLEM Schools haul away leaves and food scraps and spray their grounds, starving decomposers and harming pollinators.

SOLUTION Start a compost system and a pesticide-free grounds pledge so nutrients cycle and helpful bugs survive.

MISSION M6 Leafcutter Frenzy + M7 Humongous Fungus

CONNECTS BIOCORE ★, BIOBUZZ

PROTOTYPE A working mini-compost bin + a no-spray zone sign.

WHERE Cafeteria, school grounds

10

Sidewalk BioBlitz

PROBLEM We can't protect city nature we haven't counted — and most people think cities have none.

SOLUTION Run a neighborhood “BioBlitz,” logging species with a phone app (iNaturalist) over time.

MISSION M1 Drone Survey + M9 Research Platform

CONNECTS BIOCORE, BIOBUZZ

PROTOTYPE A BioBlitz logging sheet + a species tally chart.

WHERE Schoolyard, sidewalks, local park

Rural Canopy — 10 starters

1

Hedgerow Highway

PROBLEM Big open fields leave wildlife with no cover to travel between woodlots, so populations get cut off.

SOLUTION Plant or restore hedgerows and field-edge shrubs as wildlife corridors linking habitat patches.

MISSION M8 Tangled + M4 Lucky Leaves

CONNECTS BIOBUZZ ★, BIOCORE

PROTOTYPE Farm map + model hedgerow linking two woodlots.

WHERE Field edges, fence lines, property borders

2

Brush-Pile Havens & Nest Boxes

PROBLEM “Clean” farms remove brush, dead snags, and old barns — the shelter small animals, owls, and bats need.

SOLUTION Build brush piles and put up barn-owl and bat boxes to bring back farm helpers.

MISSION M3 Flip the Rock + M10 Fragile Microhabitats

CONNECTS BIOCORE ★, BIOBUZZ

PROTOTYPE A model brush pile + a nest/bat box.

WHERE Field corners, barns, fence lines

3

Field-Edge Wildflower Strips

PROBLEM Crop monocultures offer almost no flowers, so pollinators — and the crops that need them — suffer.

SOLUTION Plant native wildflower strips around and between fields as pollinator refuel and habitat.

MISSION M4 Lucky Leaves + M2 Exploding Seeds

CONNECTS BIOBUZZ ★

PROTOTYPE A field diagram with wildflower borders + a seed-mix plan.

WHERE Crop margins, ditch banks, unused corners

4

Cover-Crop & No-Till

PROBLEM Bare, tilled soil between crops breaks the underground fungal web and washes away.

SOLUTION Plant cover crops and reduce tilling so the soil's living web (roots + fungi) stays intact.

MISSION M5 Reaching Roots + M7 Humongous Fungus

CONNECTS BIOCORE ★

PROTOTYPE Two soil jars — tilled/bare vs. cover-cropped — showing roots + structure.

WHERE Any field, garden, or raised bed

5

Stream Buffers & Pond Edges

PROBLEM Farm runoff (soil, fertilizer) pours into creeks and ponds, choking aquatic life.

SOLUTION Plant “buffer” strips of native grasses and trees along waterways to filter runoff.

MISSION M10 Fragile Microhabitats + M5 Reaching Roots

CONNECTS BIOCORE ★

PROTOTYPE A model creek with vs. without a plant buffer (pour muddy water to test).

WHERE Farm creeks, ditches, pond edges

6

Farmyard Dark-Sky

PROBLEM All-night barn and yard lights wipe out fireflies, moths, and the bats that eat crop pests.

SOLUTION Shield and motion-switch farm lights to protect night wildlife — and get free pest control.

MISSION M7 Humongous Fungus

CONNECTS BIOBUZZ, BIOCORE

PROTOTYPE Model farmyard: glary vs. shielded lights + a firefly-count plan.

WHERE Barns, yards, driveways

7

Farm Keystone Watch

PROBLEM Farmland keystones — beavers, bats, dung beetles, barn owls — go unseen and are often removed.

SOLUTION Spotter cards + a watch log for the keystone helpers on a farm.

MISSION M13 Keystone Species + M9 Research Platform

CONNECTS BIOCORE ★

PROTOTYPE Farm keystone spotter cards + a report form.

WHERE Pastures, ponds, barns, fields

8

Restore the Back Forty

PROBLEM Marginal or worn-out field corners sit bare or invasive-choked, lost to biodiversity.

SOLUTION Replant a field corner or marginal strip with native seed to rebuild habitat.

MISSION M14 Seeds of Renewal + M2 Exploding Seeds

CONNECTS BIOCORE ★, BIOBUZZ

PROTOTYPE A restoration plan + seed balls for a real corner.

WHERE Field corners, marginal land, roadside

9

Beneficial-Bug Farming (IPM)

PROBLEM Broad pesticide spraying kills the good bugs — ladybugs, bees, decomposers — along with the pests.

SOLUTION Use Integrated Pest Management: invite beneficial insects and spot-treat instead of blanket spraying.

MISSION M6 Leafcutter Frenzy + M7 Humongous Fungus

CONNECTS BIOCORE ★, BIOBUZZ

PROTOTYPE A “good bug vs. pest” guide + an insectary-plant plan.

WHERE Crops, gardens, orchards

10

Farm BioBlitz & Pond Dip

PROBLEM Farmers rarely measure the wild biodiversity on their land, so they can't tell if it's thriving.

SOLUTION Run a farm BioBlitz — a pond dip, a night moth sheet, a bird count — logged over time.

MISSION M1 Drone Survey + M9 Research Platform

CONNECTS BIOCORE, BIOBUZZ

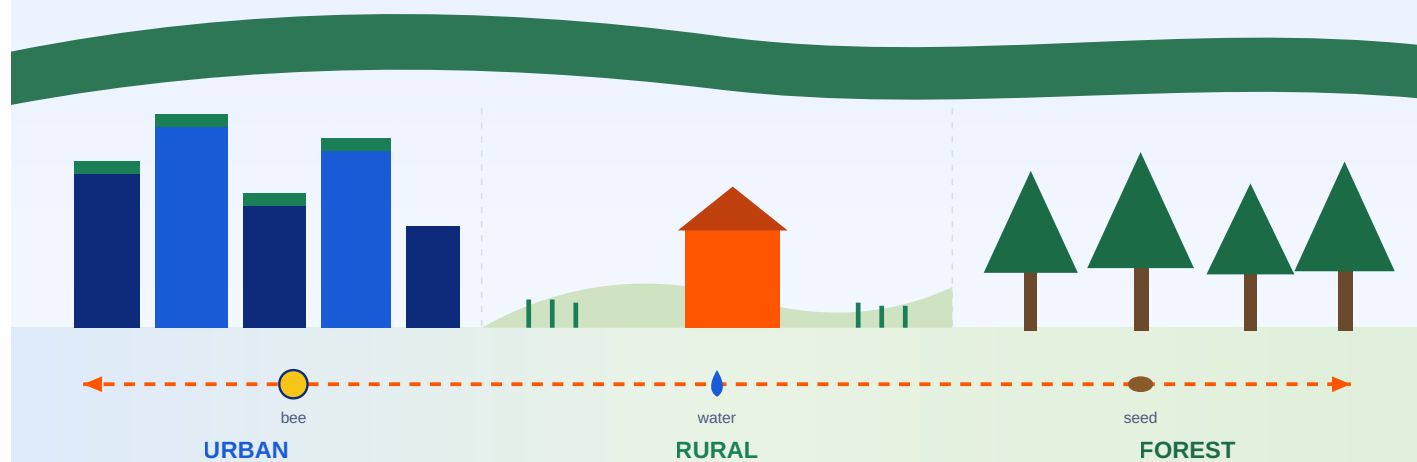
PROTOTYPE A pond-dip kit + a species tally chart.

WHERE Ponds, fields, hedgerows, night yard

Part 3 — One canopy, three tiers

City, farm, and forest aren't three worlds — they're **one continuous gradient**. A bee born in a city living wall refuels at a farm wildflower strip and pollinates a forest edge; a raindrop runs from a city rain garden to a farm stream buffer to a forest headwater; a seed rides a bird from a woodlot to a hedgerow to a vacant lot. The canopy is one unbroken web, and each setting is a link in the same chain — break any link and the chain weakens. The biggest projects **restore the connection between the tiers**: BIOGLOW = the forest link, BIOBUZZ = the pollinators that cross every link, BIOCORE = the soil-and-water web beneath them all.

One canopy — the same living web runs from the city to the forest



Cross-tier flagship projects

Standout, systems-thinking projects that follow one thing across all three tiers and mend the broken link. Full packets follow.

Follow the Bee — a pollinator highway across all three

Problem A single pollinator must feed from spring to fall by traveling across city, farm, and forest — but the flowers run out in the gaps between them. **Solution** Map one pollinator's journey city→farm→forest, find the “hungry gaps,” and design a chain of blooms (living wall → wildflower strip → forest edge) that keeps her fed all season.

Mission M4 Lucky Leaves + M8 Tangled **Connects** BIOBUZZ ★ (all tiers), BIOCORE **Prototype** Three-panel model (city/farm/forest) with a bee path and bloom “fuel stops”; mark the gaps.

1 • Identify

Pollinators need flowers from early spring to late fall, but no single place blooms the whole time. A bee may start in a city garden, cross farmland, and end at a forest edge — and if any stretch has no flowers when she needs them, she goes hungry. The connection at risk isn't one garden; it's the continuity of bloom across the whole gradient.

RESEARCH QUESTIONS

1. When do city, farm, and forest habitats bloom across the year?
2. How far can a bee travel and forage?
3. Which native plants fill the seasonal 'gap' months?

2 • Interview an expert

Who: A beekeeper, a pollinator ecologist, or a county extension agent.

ASK

1. How far do local bees travel to find food?
2. When in the year is food hardest for them to find?
3. Do city, farm, and forest bloom at different times?
4. How would you link them into one year-round food supply?
5. Where's the biggest 'hungry gap' around here?

3 • Create — prototype

MATERIALS

- Three boards for city / farm / forest scenes
- Paper or clay flowers
- A bee token on a string 'path'
- Bloom-season labels
- A map of a real route near you

BUILD STEPS

1. Build the three scenes side by side.
2. Draw the bee's path across all three.
3. Add bloom 'fuel stops' and label each one's bloom season.
4. Find a stretch/month with no flowers — the hungry gap.
5. Add plantings that bloom then, and re-check the route.

4 • Iterate & 5 • Communicate

Lay out bloom times across the year for your three scenes; find the weeks with no flowers anywhere on the route, add plants that bloom then, and count how many gap-weeks you closed.

Measure: Number of 'no-bloom' weeks along the route, before vs. after your plantings.

“Meet Beatrice the bee — she has to eat from March to October, but the city, the farm, and the forest each feed her for only part of the year. Watch what happens in the hungry months.”

Follow the Raindrop — one watershed, three tiers

Problem Rain that falls on a city runs to a farm creek and on to a forest stream, carrying pollution downstream and harming life in every tier. **Solution** Trace one raindrop's path through your watershed (city → farm → forest waters) and design the rain gardens and buffers that clean and slow it at each stage.

Mission M10 Fragile Microhabitats + M5 Reaching Roots **Connects** BIOCORE ★, BIOBUZZ **Prototype** A tilted 3-zone model watershed you can pour water through, with buffers you add and re-test.

1 • Identify

Water connects everything downhill. A storm-drain in the city, a ditch on a farm, and a forest stream are the same water at different points on its journey. Trash and fertilizer picked up in one tier poison life in the next. The connection at risk is the health of the water as it flows across all three tiers — you can't fix it in just one place.

RESEARCH QUESTIONS

1. What is a watershed, and where does our water actually go?
2. What pollutants does stormwater runoff pick up?
3. How do rain gardens and plant buffers clean and slow water?

2 • Interview an expert

Who: A creek/watershed group volunteer, a soil & water conservation district staffer, or a hydrologist.

ASK

1. Where does the rain that hits our school end up?
2. What gets into the water along the way?
3. How do buffers and rain gardens help?
4. Which stretch of our watershed needs help most?
5. How would we know if the water is getting cleaner?

3 • Create — prototype

MATERIALS

- A tilted tray/board with 3 zones
- Soil + cocoa or food coloring as 'pollution'
- Sponge/plant props for buffers
- Cups to catch the runoff
- A watering can

BUILD STEPS

1. Build a sloped model with city, farm, and forest zones.
2. Sprinkle 'pollution' across the zones.
3. Pour 'rain' and catch the runoff at the bottom — it's dirty.
4. Add a rain garden, a farm buffer strip, and a forest wetland.
5. Pour again and compare how clean the water comes out.

4 • Iterate & 5 • Communicate

Compare the clarity and amount of runoff caught at the 'forest' end before vs. after adding a buffer at each tier.

Measure: Runoff clarity/volume at the outlet, before vs. after buffers.

"Follow one raindrop from our parking lot — in four minutes it'll be in a stream miles away. Let's see what it drags along with it."

Follow the Seed — regeneration across the gradient

Problem Forests and meadows regrow when seeds travel, but seeds carried by birds and wind can't cross the long dead zones between city, farm, and forest. **Solution** Track how a seed moves across the gradient and add 'stepping-stone' habitat (a vacant lot, a field corner, a woodlot) so seeds and their animal carriers can hop city→farm→forest.

Mission M2 Exploding Seeds + M14 Seeds of Renewal **Connects** BIOCORE ★, BIOBUZZ **Prototype** A 3-zone map showing a seed's hops, a gap too wide to cross, and stepping-stone patches that complete the route.

1 • Identify

New plants appear where seeds land, and most seeds hitch a ride on birds, mammals, or wind. Between a forest and a city there's often farmland or pavement with nowhere for a seed-carrier to stop, so seeds never make the trip and habitats can't spread or recover. The connection at risk is the chain of stepping-stone habitats that lets regeneration move across the whole gradient.

RESEARCH QUESTIONS

1. How do seeds travel, and how far?
2. Which animals carry seeds between habitats?
3. How close do 'stepping-stone' habitats need to be?

2 • Interview an expert

Who: A restoration ecologist, a land-trust volunteer, or a naturalist.

ASK

1. How do new plants show up in an empty lot or field?
2. What animals move seeds around here?
3. How far can a seed realistically travel?
4. Where are the gaps too big to cross?
5. What small patches would help the most?

3 • Create — prototype

MATERIALS

- A 3-zone map board
- Seed and bird tokens
- Patch cut-outs for stepping stones
- Distance markers/ruler

BUILD STEPS

1. Map city, farm, and forest with the real distances.
2. Place a bird/seed and hop it toward the forest.
3. Find where the gap is too far to cross.
4. Add stepping-stone patches (a lot, a field corner, a woodlot).
5. Re-run the hops and complete the route.

4 • Iterate & 5 • Communicate

Measure the biggest gap a seed-carrier must cross before vs. after adding stepping-stone patches, and show the route now connects.

Measure: Largest gap distance along the route, before vs. after adding patches.

"This acorn wants to become a forest — but first it has to get past a highway, a cornfield, and a parking lot. Let's help it hop."

The Corridor Challenge — map & mend a real route

Problem The nearest wild land may be closer than we think, but the path to it is broken by roads, lawns, and pavement that wildlife can't cross. **Solution** Map the real route from your school to the nearest natural area, find every gap in the 'green chain,' and design fixes — a corridor, a crossing, a set of stepping-stones — to reconnect it.

Mission M8 Tangled + M1 Drone Survey **Connects** BIOCORE ★, BIOBUZZ **Prototype** A real map from school to the nearest wild land, marked with green links and red gaps, plus proposed fixes.

1 • Identify

Habitat only works if it's connected. Around most schools there are scattered green patches — yards, a park, a creek, a woodlot — but the links between them are broken by roads and pavement, so wildlife is trapped on islands. The connection at risk is the continuous corridor from where the kids are to the nearest wild place; mapping it makes the gaps (and the fixes) obvious.

RESEARCH QUESTIONS

1. Where is the nearest natural area to our school?
2. What blocks wildlife from moving there?
3. What makes a good wildlife corridor or crossing?

2 • Interview an expert

Who: A city/greenway planner, a parks department staffer, or a GIS/mapping specialist.

ASK

1. What green spaces are near our school?
2. What stops animals from moving between them?
3. Are there corridors or crossings in our area already?
4. Which gap would be easiest to fix first?
5. Who decides where green space goes?

3 • Create — prototype

MATERIALS

- A printed or online map
- Markers (green links, red gaps)
- Tracing paper
- A ruler / map scale
- Photos of the gaps

BUILD STEPS

1. Mark your school and the nearest wild land on a map.
2. Draw the green patches in between.
3. Connect them and mark every break as a red gap.
4. Pick the most fixable gap.
5. Design a corridor, crossing, or stepping-stone fix for it.

4 • Iterate & 5 • Communicate

Count the number of gaps (breaks in the green chain) along the route, then show how your proposed fixes reduce that count.

Measure: Number of gaps in the green chain, before vs. after your proposed fixes.

"The nearest forest to our school is only a few blocks away — but a squirrel could never make it. Here's the map of everything in its way."

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